

Chapter 4: General Dynamic Games

ECON 5001

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Based on: *Game Theory* by Giacomo Bonanno

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What Chapter 3 Assumed

Last time, every player knew the entire history when it was her turn.

That ruled out almost everything interesting:

- sealed bids
- a firm choosing capacity without seeing its rival's choice
- poker
- any move made **at the same time** as somebody else's

In this lecture we drop that assumption, then rebuild backward induction so it still works.

Two Columbus Businesses

Plenty of Columbus taprooms have no kitchen. The food comes from a truck parked outside.

- **The Brewery** has one taproom and is thinking about growing
- **The Truck** wants a season-long spot outside a busy taproom

The Brewery decides first:

- **Stay:** expand the current taproom. Safe, predictable.
- **Expand:** open a second taproom, either in **Franklinton** or in the **Short North**

If the Brewery expands, the Truck picks its home for the season: **Franklinton** or the **Short North**.

Hypothetical firms, real neighborhoods.

What Each One Wants

- Both want to land in the **same** neighborhood. A new taproom with no food, or a truck with no taproom, does badly.
- The Brewery prefers **Franklinton**: cheaper rent, room for a patio
- The Truck prefers the **Short North**: late-night foot traffic on High Street

	Brewery	Truck
Brewery stays	2	0
Both in Franklinton	3	1
Both in the Short North	1	3
Different neighborhoods	0	0

Extra annual profit, in \$100,000s. The Truck's 0 is just an ordinary season.

Once the Brewery expands, this is Battle of the Sexes.

Who Knows What

The Brewery posts on Instagram: “*Second taproom coming next spring!*”

But it keeps the address secret until the lease is signed, and the Truck has to lock in its spot that same week.

- ✓ The Truck **sees** whether the Brewery chose Stay or Expand
- ✗ The Truck does **not see** which neighborhood the Brewery chose

The Truck knows *whether*. It doesn't know *where*.

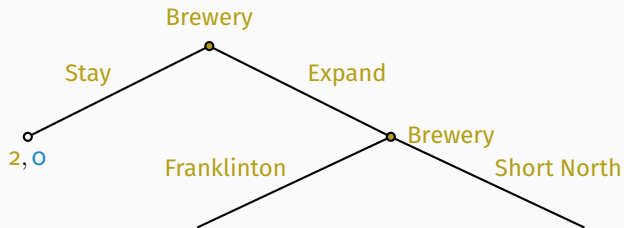
Neither of our tools fits. Chapter 3 assumed the Truck sees everything. A simultaneous game would assume it sees nothing.

Drawing It



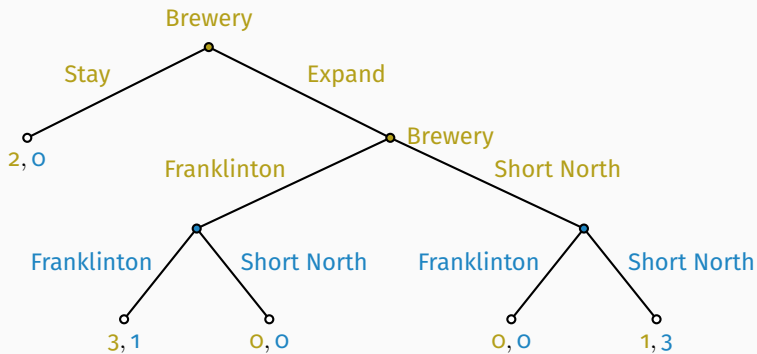
The Brewery moves first. Stay ends the game.

Drawing It



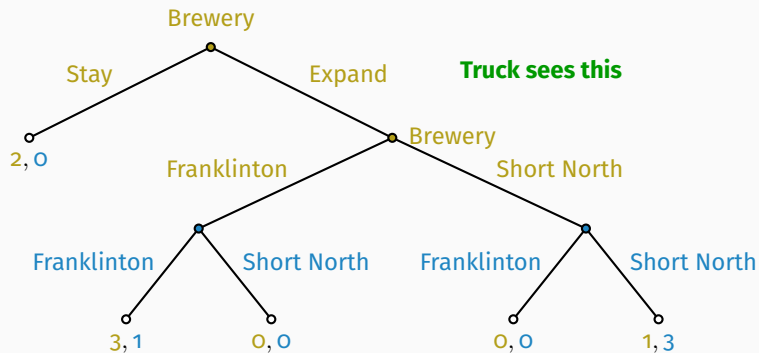
After Expand, the Brewery picks a neighborhood.

Drawing It



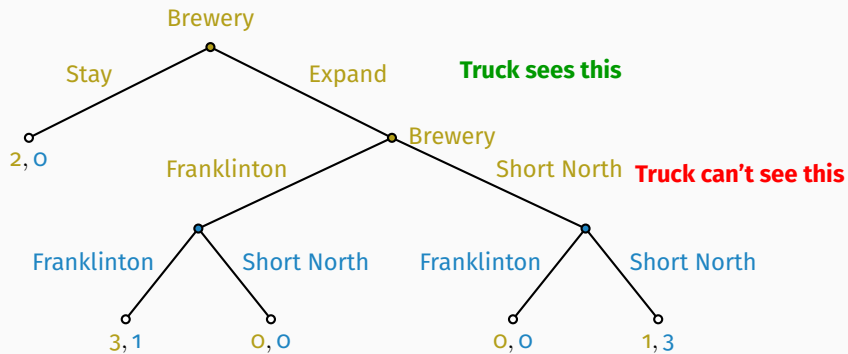
Then the Truck picks. So far this looks like a Chapter 3 tree.

Drawing It



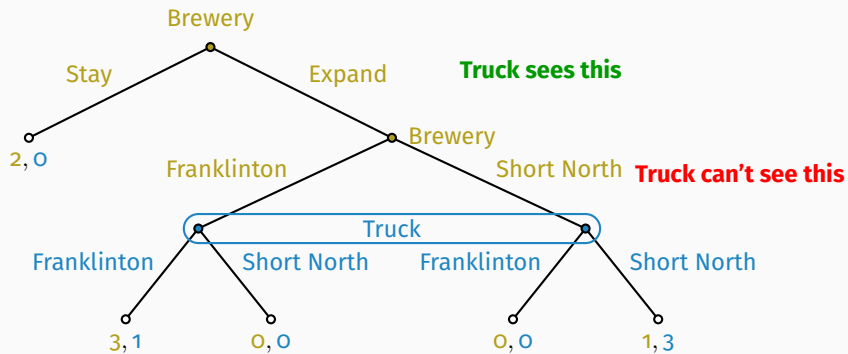
The Truck observes the Stay/Expand decision. Every Truck node comes after Expand.

Drawing It



But it does not observe the neighborhood decision.

Drawing It



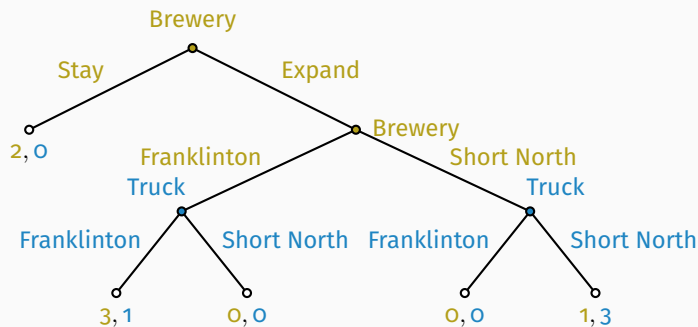
The oval: the Truck knows it is at **one of these two nodes**, but not which one.

From the Truck's Seat

- If the Brewery stays, the Truck never moves. Nothing to decide.
- If the Brewery expands, the Truck is somewhere **inside the oval**. It knows it is past Expand. It doesn't know which branch came after.
- So it picks **one** neighborhood for the whole oval. A plan like "*Franklinton if they chose Franklinton*" is not available. It depends on something the Truck can't see.
- What the Truck *did* see still tells it something: the Brewery passed up a sure 2.

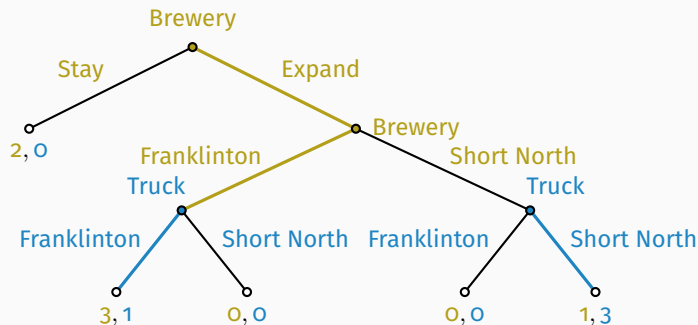
This oval has a name, and it is the main new idea of this chapter.

What If the Truck Could See Everything?



Suppose the address leaked. No oval: two separate Truck nodes, and we are back in Chapter 3.

What If the Truck Could See Everything?



Suppose the address leaked. No oval: two separate Truck nodes, and we are back in Chapter 3.

Backward induction: the Truck follows the Brewery wherever it goes. Knowing that, the Brewery expands to Franklinton and gets 3. Outcome (3, 1).

Somewhere in Between

	Sees Stay vs. Expand?	Sees the neighborhood?
Chapter 3 (perfect information)	✓	✓
Simultaneous-move game	✗	✗
Our game	✓	✗

Real decisions are usually in the last row. You see *some* of what others did, not all of it.

Because the Truck sees part of the history, this is **not** just a simultaneous 2×2 game, and we can't solve it by backward induction alone. We need new tools.

Think About It

You own the Truck. You just saw *"Second taproom coming next spring!"*

Franklinton or the Short North?

Hint: the Brewery could have stayed put and earned 2 for sure. What does *giving that up* tell you about where it's going?

Information Sets

A Familiar Situation

To discourage copying, I print two versions of the exam, white paper and pink, and hand them out so neighbors never share one.

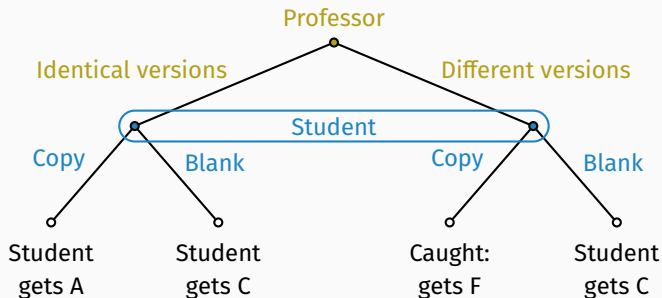
Or do I? Writing genuinely different versions is extra work for me. I might be bluffing: same questions, different paper.

A student who didn't study is sitting next to someone who did.

- Turn in a blank exam and, given earlier grades, he gets a **C**
- Copy, and if the versions are identical he gets an **A**
- Copy, and if the versions differ he is caught and gets an **F**

The moves are sequential. I decide first, but he doesn't see my decision.

Drawing What He Doesn't Know



The enclosure is an **information set**: the Student knows he is at one of those two nodes, but not which one.

Definition: Extensive-Form Game-Frame

Everything from Chapter 3, plus one item:

- a finite rooted directed tree
- players $I = \{1, \dots, n\}$, one assigned to every decision node
- actions on edges, no two edges out of a node sharing an action
- outcomes assigned to terminal nodes

★ For every player i , a **partition** $\mathcal{D}_i$ of her decision nodes D_i . Each cell of the partition is an **information set**.

We write H for a single information set, so $H \in \mathcal{D}_i$.

Restriction: any two nodes in the same information set must offer *exactly the same actions*.

Why must that hold?

A Note on Notation

Three different things, easily confused:

- D_i is the set of **all** decision nodes belonging to player i
- $\mathcal{D}_i$ is the **partition** of D_i , a collection of sets
- $H \in \mathcal{D}_i$ is **one** information set, a single cell

Bonanno writes D and D' where we write H and H' . Watch out: in Chapter 3 he also used D for the set of *all* decision nodes in the tree, so the same letter does two jobs. We'll stick with H for information sets.

Perfect Information Is the Special Case

Every information set is a singleton



Perfect information

So Chapter 3 was not a different theory. It was this one, with the partition as fine as it can be.

Convention from here: we draw an enclosure **only** around information sets with two or more nodes. A lone node is its own information set.

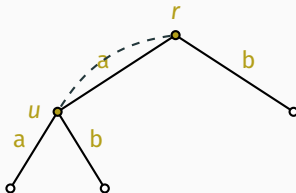
Perfect Recall: First Condition

Almost all of game theory assumes players don't forget.

Condition 1

No node in an information set is a **predecessor** of another node in it.

Below, $H = \{r, u\}$ violates it: r comes before u , so Player 1 cannot tell whether she has already moved.

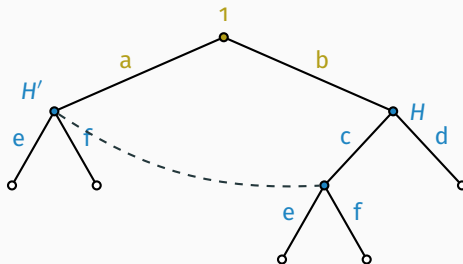


Perfect Recall: Second Condition

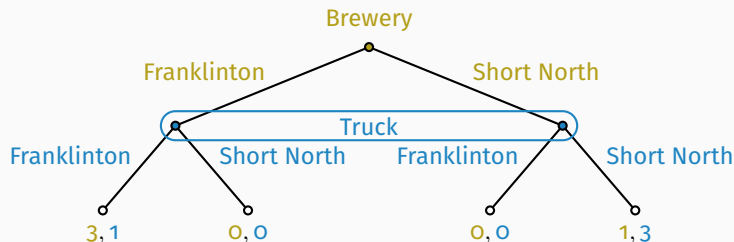
Condition 2

If $x \in H$ leads to $y \in H'$ by action a , then **every** node in H' has a predecessor in H reached by that same a .

Below, H' is Player 2's lower information set. Its right node is reached via H , but its left node never passes through H at all. Player 2 cannot tell whether this is her first move or her second.

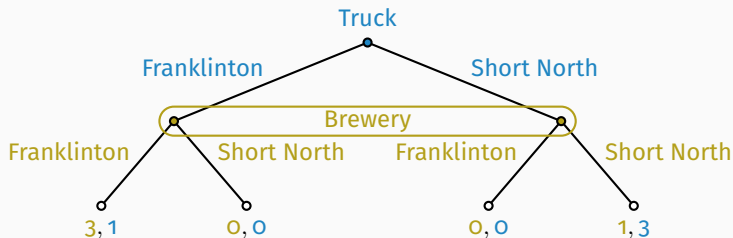


The Neighborhood Stage, Drawn Two Ways



Cut off the Stay/Expand move and keep what follows Expand. The Brewery is drawn first, but the Truck's information set means it sees nothing. So this piece is just the simultaneous 2×2 Battle of the Sexes.

The Neighborhood Stage, Drawn Two Ways



Cut off the Stay/Expand move and keep what follows Expand. The Brewery is drawn first, but the Truck's information set means it sees nothing. So this piece is just the simultaneous 2×2 Battle of the Sexes.

Swap who is drawn first and nothing changes. Simultaneity is not about time. It is about **what you know when you choose**.

Strategies, Again

One Choice per Information Set

Definition

A **strategy** for a player in an extensive-form game is a list of choices, one for every **information set** of that player.

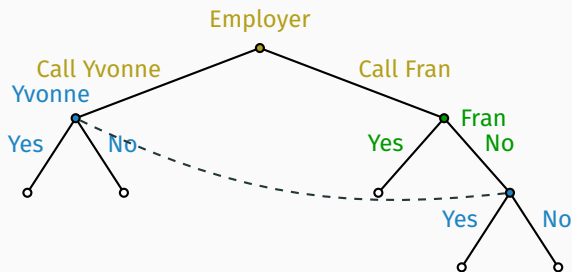
Chapter 3 said “one for every decision node.” That was the same rule, because every node was its own information set.

The change matters: a player can no longer make a plan that depends on something she cannot observe.

A Plan You Are Not Allowed to Make

Two candidates, Yvonne and Fran, interview for one position. The employer says: “don’t call me, I’ll call you.” If the first candidate turns him down, he immediately calls the other, never saying it’s a second offer.

Yvonne would love to plan: *If he calls me first, I say Yes. If I’m the recycled offer, I say No.*



Must choose the **same action at every node** of an information set.

From Tree to Table, Again

Every strategy profile still determines a path, hence an outcome, hence payoffs. So every extensive-form game still has a **strategic form**.

And we could just find its Nash equilibria and stop.

But Chapter 3 taught us not to. Some Nash equilibria of the strategic form rest on threats nobody would carry out.

We fixed that with backward induction, which needs a *last* decision node.

What is the analogue when players move blind?

Subgames

The Idea

A **subgame** is a piece of the tree that could stand alone as a game.

Informal procedure:

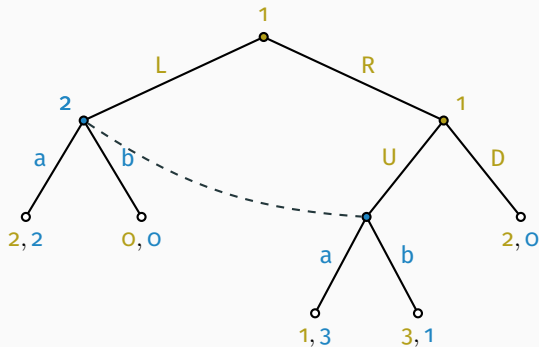
1. Pick a decision node x whose information set is **just** $\{x\}$
2. Draw an oval around x and everything that follows it

If that oval **cuts through** an information set, the oval is illegal. So only call it a subgame if it cuts **no** information set.

“Cutting” means some information set has one node inside your oval and another node outside it.

Every game is a subgame of itself. Anything smaller is a **proper** subgame.

Why Cutting Is Fatal



Player 1's node after *R* looks like a fine place to start. But the oval around it would contain *one* of Player 2's two nodes.

Player 2 can't tell those nodes apart, so you cannot hand her the right-hand piece and call it a game. Half her uncertainty is outside it.

Definition: Subgame

Let $S(x)$ be the set containing x and all its successors.

$S(x)$ is a **subgame starting at x** if:

1. the information set containing x is the singleton $\{x\}$, and
2. for every information set H in the game, if $H \cap S(x) \neq \emptyset$ then $H \subset S(x)$

It is a **proper** subgame if x is not the root.

A subgame is **minimal** if it contains no other subgame strictly inside it. Those are the ones we solve first.

Subgame-Perfect Equilibrium

Definition

Let s be a strategy profile for the whole game, and G a subgame. Write $s|_G$ for the **restriction** of s to G : the part of s that prescribes choices at G 's information sets, and only those.

Definition

s is a **subgame-perfect equilibrium** if, for **every** subgame G , the restriction $s|_G$ is a Nash equilibrium of G .

The whole game is one of its own subgames, so every subgame-perfect equilibrium is a Nash equilibrium.

Not every Nash equilibrium is subgame perfect. That's the point.

IKEA and Wayfair, Again

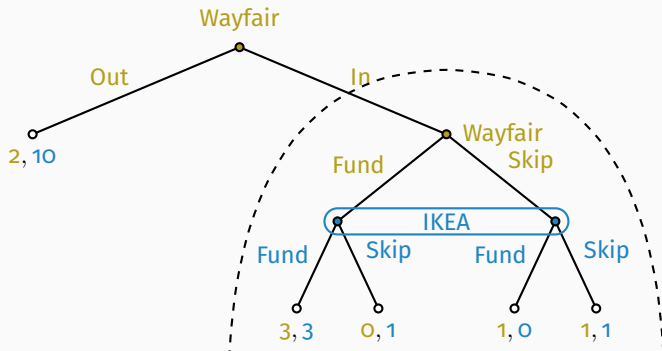
IKEA is the established furniture store at Polaris, earning \$10 million a year in profit. Wayfair decides whether to open across I-71.

If Wayfair opens, the two stores **simultaneously** decide whether to fund a joint campaign advertising Polaris as a furniture destination.

- Both fund it: shoppers drive up from all over. Both gain.
- Neither funds it: nothing changes.
- One funds it alone: too small to work, and the money is gone.

That second stage is a coordination game too. Unlike the Brewery and the Truck, both stores rank its two equilibria the same way.

The Whole Game



Annual profit in \$millions. First number Wayfair, second IKEA. The circle marks the only proper subgame.

The Algorithm

1. Find a **minimal** subgame.
2. Solve it. Find its Nash equilibria, pick one, and note the strategies.
3. **Mark** the subgame's starting node with that equilibrium's payoff vector, exactly as in backward induction, then delete everything below it. You now have a smaller game.
4. Repeat until no proper subgames remain. Solve what's left. Patch together everything you noted along the way.

Same marking idea as Chapter 3. The only change: a minimal subgame may be a whole simultaneous game rather than a single node, so we mark it with a Nash equilibrium payoff instead of one player's best choice.

If a subgame had several Nash equilibria, run the procedure again with a different one for a different subgame-perfect equilibrium.

If some subgame has *no* Nash equilibrium, neither does the game.

Step 1: Solve the Subgame

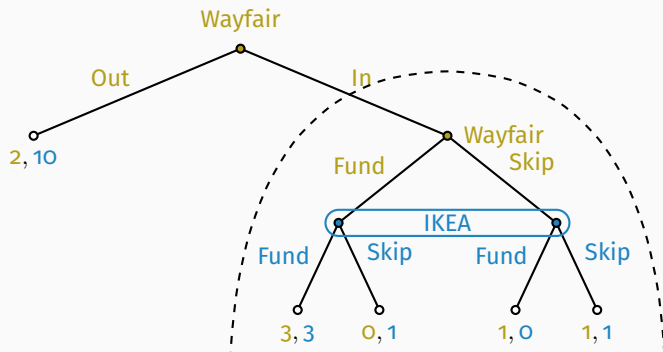
		IKEA	
		Fund	Skip
Wayfair	Fund	<u>3</u> , <u>3</u>	0, 1
	Skip	1, 0	<u>1</u> , <u>1</u>

Two Nash equilibria: (Fund, Fund) with payoffs (3, 3), and (Skip, Skip) with payoffs (1, 1).

So we have to run the algorithm **twice**.

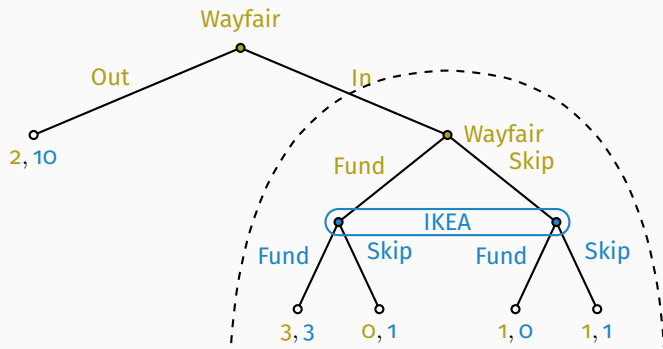
Step 2: Replace and Solve

Using the **(Fund, Fund)** equilibrium.



Step 2: Replace and Solve

Using the **(Skip, Skip)** equilibrium.



Two Subgame-Perfect Equilibria

$((\text{In}, \text{Fund}), \text{Fund})$ and $((\text{Out}, \text{Skip}), \text{Skip})$

In the second one Wayfair never opens, *because* it expects no campaign.

Nothing about the market changed. The pessimism is self-confirming.

- Both firms would prefer the first equilibrium
- Neither can get there by acting alone
- Subgame perfection does **not** pick between them

Refinements narrow the set of predictions. Not always to one.

The Strategic Form

Wayfair's strategy is a pair: what to do at the root, and what to do in the campaign stage.

		IKEA	
		Fund	Skip
Wayfair	(Out Fund)	2, 10	2, 10
	(Out Skip)	2, 10	2, 10
	(In Fund)	3, 3	0, 1
	(In Skip)	1, 0	1, 1

The Strategic Form

Wayfair's strategy is a pair: what to do at the root, and what to do in the campaign stage.

		IKEA	
		Fund	Skip
Wayfair	(Out Fund)	2, <u>10</u>	<u>2</u> , <u>10</u>
	(Out Skip)	2, <u>10</u>	<u>2</u> , <u>10</u>
	(In Fund)	<u>3</u> , <u>3</u>	0, 1
	(In Skip)	1, 0	1, <u>1</u>

Three cells have both payoffs underlined. Three Nash equilibria.

The Nash Equilibrium We Threw Out

The strategic form has **three** Nash equilibria. Wayfair's strategy is a pair: what to do at the root, and what to do in the campaign stage.

Profile	Subgame play	Subgame perfect?
((In, Fund), Fund)	(Fund, Fund)	yes
((Out, Skip), Skip)	(Skip, Skip)	yes
((Out, Fund), Skip)	(Fund, Skip)	no

The third is a Nash equilibrium of the whole game. Wayfair staying out is a best reply if it believes IKEA will skip.

But its plan for the subgame is incoherent: if entry happened, funding alone earns 0, and it would rather skip.

Three Things Worth Remembering

- In a **perfect-information** game, subgame-perfect equilibrium is exactly backward induction. Every decision node starts a subgame, so the algorithm walks the tree from the leaves.
- If a game has **no proper subgames**, every Nash equilibrium is subgame perfect. The refinement has nothing to bite on.
- Otherwise subgame perfection is a strict refinement: fewer predictions, all of them Nash.

Bonanno's Figures 4.8 and 4.9 work through larger three-player trees with several subgames. Worth doing before the problem set.

Chance Moves

When Nobody Chose It

Sometimes what happens isn't anyone's decision. A card is dealt, a part fails, a customer walks in.

We model this with a fictitious player: **Nature** (or **Chance**).

- Nature moves at some nodes
- Each of Nature's actions carries a **probability**
- Nature has no payoffs and no objectives, so it isn't really a player

Everything else (information sets, strategies, subgames) is unchanged.

Three Cards

Three cards: one black, two red. Shuffled, face down.

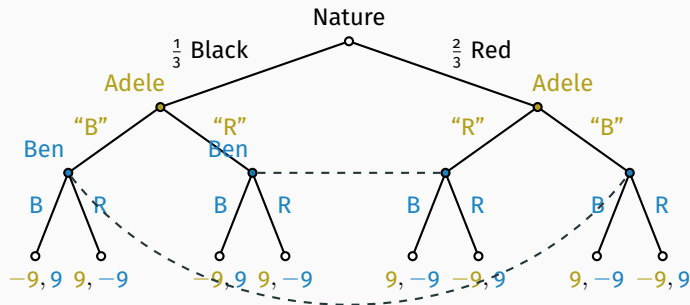
Nature moves first, turning up the top card: black with probability $\frac{1}{3}$, red with probability $\frac{2}{3}$.

Then the players move:

- Adele looks at the top card without showing Ben
- She tells Ben “black” or “red”, truthfully or not
- Ben then guesses the true color
- Correct guess: Adele pays Ben \$9. Wrong: Ben pays Adele \$9.

Ben has two information sets, one for each thing Adele might say, and in each he is unsure what the card actually is.

The Card Game



Nature moves first. Adele's label is what she says; Ben's is what he *guesses*. First payoff Adele, second Ben.

Ben has two information sets, one for each thing Adele might say. In each, he knows what he heard but not what the card is.

A New Difficulty

Take the profile: Adele always tells the truth, Ben always guesses black.

What's the outcome? It depends on the card, so the outcome is itself a gamble:

Outcome	Adele pays \$9	Ben pays \$9
Probability	$\frac{1}{3}$	$\frac{2}{3}$

We call this a **lottery**. To build a strategic form we need to know how players rank *lotteries*, not just outcomes.

Chapter 5 answers that properly. For now, one shortcut.

Expected Value

Definition

The **expected value** of the money lottery $\begin{pmatrix} \$x_1 & \$x_2 & \cdots & \$x_n \\ p_1 & p_2 & \cdots & p_n \end{pmatrix}$ is

$$x_1p_1 + x_2p_2 + \cdots + x_np_n$$

$\begin{pmatrix} \$5 & \$15 & \$25 \\ \frac{1}{5} & \frac{2}{5} & \frac{2}{5} \end{pmatrix}$ has expected value _____

$\begin{pmatrix} -\$9 & \$9 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$ has expected value _____

Risk Neutrality

Definition

A player is **risk neutral** if she considers a money lottery to be *just as good* as its expected value.

So a risk-neutral player ranks lotteries by their expected value, and we can use that number as her payoff.

$$L_1 = \begin{pmatrix} \$5 & \$15 & \$25 \\ \frac{1}{5} & \frac{2}{5} & \frac{2}{5} \end{pmatrix} \quad L_2 = \begin{pmatrix} \$16 \\ 1 \end{pmatrix} \quad L_3 = \begin{pmatrix} \$0 & \$32 & \$48 \\ \frac{5}{8} & \frac{1}{8} & \frac{1}{4} \end{pmatrix}$$

A risk-neutral player ranks these: _____

This is a convenience, not a claim about people. Risk neutrality lets us finish Chapter 4 without the machinery of Chapter 5.

Why the Assumption Is Doing Work

L_2 and L_3 have the same expected value, \$16.

- L_2 pays \$16 for certain
- L_3 pays nothing five times out of eight, and sometimes \$48

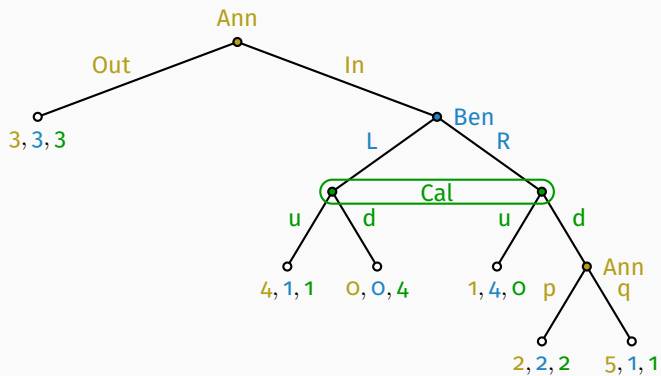
A risk-neutral player is **indifferent** between them.

Most people are not. Ask yourself which you'd take, and notice that your answer is information the expected value throws away.

Recovering it is exactly what Chapter 5 is for.

Let's Work One Out

A Three-Player Dynamic Game



A Three-Player Dynamic Game

Its strategic form. Ann's strategy is a pair: the root, then her later node.

		Ben				Ben	
		L	R			L	R
Ann	(Out p)	3, 3, 3	3, 3, 3	(Out p)	3, 3, 3	3, 3, 3	
	(Out q)	3, 3, 3	3, 3, 3	(Out q)	3, 3, 3	3, 3, 3	
	(In p)	4, 1, 1	1, 4, 0	(In p)	0, 0, 4	2, 2, 2	
	(In q)	4, 1, 1	1, 4, 0	(In q)	0, 0, 4	5, 1, 1	
		Cal: u				Cal: d	

A Three-Player Dynamic Game

1. List every subgame. Why is there none starting at Cal's left node?
2. How many strategies does each player have?
3. Run the subgame-perfect equilibrium algorithm. Solve the smallest subgame first.
4. Find a Nash equilibrium of the whole game that is **not** subgame perfect.

What Have We Learned, Charlie Brown?

- An **information set** collects nodes a player cannot tell apart; perfect information is the case where all are singletons
- A **strategy** gives one choice per information set, not per node
- A **subgame** starts at a singleton information set and cuts no information set
- **Subgame-perfect equilibrium**: Nash in every subgame. Solve minimal subgames first, replace, repeat
- **Nature** handles chance; risk neutrality ranks lotteries by mean

Next: what if players aren't risk neutral?

- Bonanno, *Game Theory*, Chapter 4
- Exercises: §4.6.1 (imperfect information), §4.6.2 (strategies), §4.6.3 (subgames), §4.6.4 (subgame-perfect equilibrium), §4.6.5 (chance moves)
- Exercise 4.2 is the entry game with a simultaneous second stage, close to the IKEA example, with more to it
- Solutions in §4.7